it has to be true that many of his pairs of genes coalesce back in Victoria. And the same applies to pairs of your genes, and pairs of my genes. Even though we may not have Prince Charles' well-documented pedigree to consult, any pair of your genes could, in principle, look back at their common ancestor, the coalescence point at which they were 'peeled off' from the same parent gene.

Now, here's something interesting. Although I can't establish the exact coalescence point of any particular allelic pair of my genes, geneticists can in principle take all the pairs of genes from any one individual and, by considering all possible pathways back through the past (actually not all possible pathways because there are too many, but a statistical sample of them), derive a pattern of coalescences across the whole genome. Heng Li and Richard Durbin of the Sanger Institute in Cambridge realized a remarkable thing: the pattern of coalescences among pairs of genes in the genome of a single individual gives us enough information to reconstruct demographic details about datable moments in the prehistory of an entire species.

In our discussion of coalescence between pairs of genes, one from the father and one from the mother, the word 'gene' means something a bit more fluid than the normal usage of molecular biologists. Indeed, you could say that the coalescence geneticists have reverted to something a bit like my 'slightly selfish big bit of chromosome and even more selfish little bit of chromosome.' Coalescence analysis is studying chunks of DNA which might be larger or even smaller than a molecular biologist's understanding of a single gene but which can still be seen as cousins of each other, having been 'peeled off' from a shared ancestor some definite number of generations ago.

When a gene (in that sense) 'peels off' two copies of itself and gives one to each of two offspring, the descendants of those two copies may, over time, accumulate differences due to mutation. These may be 'under the radar' in the sense that they don't show up as phenotypic differences. The mutated differences between them are proportional to the time that has elapsed since the split, a fact that biologists make good use of, over much greater time spans, in the so-called 'molecular clock'. Moreover, the pairs of genes whose cousinship we are calculating needn't have the same phenotypic effects as each other. I have one blue eye gene from my father paired with one brown eye gene from my mother. Although these genes are different, even they must have a coalescence somewhere in the